

Project Title
Intelligent Student Record System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by

Kairav Vashi (2520030582)

Sai Harshith Reddy (2520030375)

A. Yoshik (2520030574)

Team Number: 1

Section: 9

Under the guidance of

Dr. S. Vinay Kumar

Professor, Department of CSE



Hyderabad-500090, Telangana, India.

ABSTRACT

Educational institutions maintain extensive student records, including names, roll numbers, departments, courses, marks, attendance, and achievements. Manually searching through these records is slow, error-prone, and does not scale as the number of students grows. This project presents the **Intelligent Student Record Search System**, a Java-based application designed to store and retrieve student information efficiently using core data structures.

Records are maintained in an **ArrayList** as the primary repository, while a **HashMap** indexed by roll number enables constant-time lookup for exact matches.

The system begins with basic keyword search across name, roll number, and course fields, and progressively incorporates string matching algorithms to support partial and approximate queries. Planned enhancements include fuzzy search for handling typographical errors, relevance-based ranking of search results, and auto-complete suggestions. The project demonstrates the practical application of fundamental data structures and algorithms including searching, hashing, and string matching within a real-world, explainable use case suited for academic evaluation.

Signature of the Guide